

Beyond Linear Models

DS 4440 · LECTURE 3

1. Last time: linear regression from an “ML” perspective

That is, via $\nabla \mathcal{L}$ and SGD. The gradient of squared error was

$$\nabla_{\mathbf{w}} \mathcal{L} = \underbrace{-2\mathbf{X}^\top}_{d \times n} \underbrace{\mathbf{y}}_{n \times 1} + 2 \underbrace{\mathbf{X}^\top \mathbf{X}}_{d \times d} \underbrace{\mathbf{w}}_{d \times 1}.$$

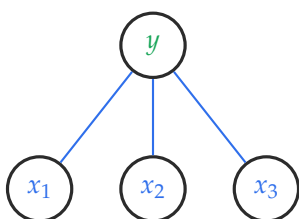
Analytic solution

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Gradient descent

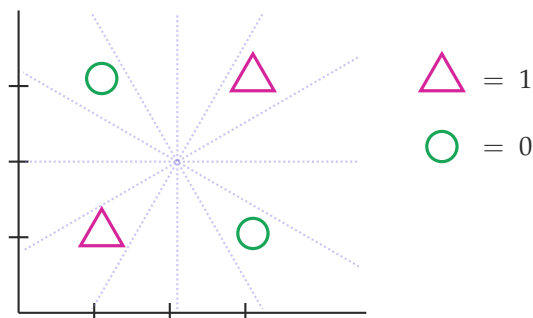
$$\hat{\mathbf{w}}^{(t+1)} \leftarrow \hat{\mathbf{w}}^{(t)} - \underbrace{\alpha}_{\text{learning rate}} \nabla_{\mathbf{w}} \mathcal{L} \left(\underbrace{\mathbf{X}, \mathbf{y}}_{\text{usually batched}} \mid \hat{\mathbf{w}}^{(t)} \right)$$

2. So far we have assumed...



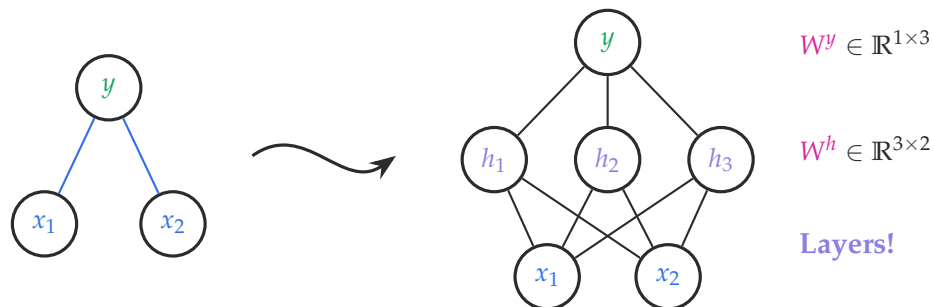
$$y = f(w_0 + w_1 x_1 + w_2 x_2 + w_3 x_3)$$

This implies a line or **hyperplane** that separates classes. But consider:



How can we address this?

3. Add a layer



The hidden layer is just a matrix multiply: a 3×2 weight matrix applied to the 2×1 input gives three hidden values.

$$\begin{array}{c} 2 \\ \hline \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array} \begin{array}{l} h_1 \\ h_2 \\ h_3 \end{array} \end{array} \quad \begin{array}{c} \begin{array}{|c|} \hline \\ \hline \end{array} \begin{array}{l} x_1 \\ x_2 \end{array} \end{array} \quad \boxed{h = W^h x}$$

But this is still linear!

Write $W^h_{j:}$ for row j of W^h , so that $h_j = W^h_{j:} \cdot x$. Then

$$\begin{aligned} y &= W^y_1 \cdot h_1 + W^y_2 \cdot h_2 + W^y_3 \cdot h_3 \\ &= W^y_1(W^h_{1:} \cdot x) + W^y_2(W^h_{2:} \cdot x) + W^y_3(W^h_{3:} \cdot x). \end{aligned}$$

So this is just a weirdly parameterized **linear model**. (See the Colab notebook.)

4. Quick aside: binary cross entropy (BCE)

$$\mathcal{L} \stackrel{\text{def}}{=} -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

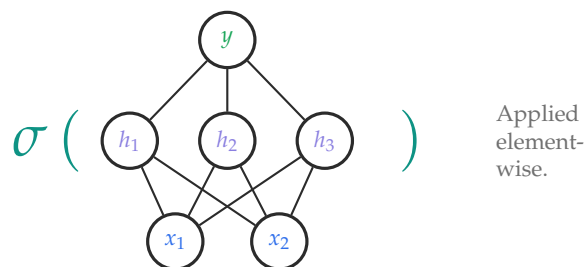
$$y \in \{0, 1\} \quad \cdot \quad \hat{y} = \text{predicted probability}$$

We want the bracketed term to be big — i.e. high $\hat{y}$ when $y = 1$, and small otherwise.

(As in your homework!)

Only one of the two terms is ever active: if $y = 1$ the loss is $-\log \hat{y}$, and if $y = 0$ it is $-\log(1 - \hat{y})$. Either way, confidently predicting the wrong label drives the loss toward infinity.

5. Enter activation functions



The fix is to put a nonlinear σ between the layers:

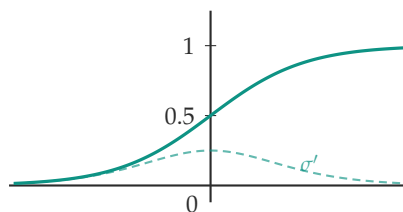
$$h = \sigma(W^h x), \quad y = W^y h.$$

Now $W^y W^h$ cannot be collapsed into a single matrix, and the model can carve out non-linear decision boundaries.

6. One choice for σ : sigmoid

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

$$\frac{d}{dx}\sigma(x) = \sigma(x)(1 - \sigma(x))$$



Note: SoftMax $\neq$ Sigmoid!

$$\text{SM}(\underbrace{z}_{\text{a vector}})_i = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Sigmoid squashes one scalar independently; softmax takes a whole vector of scores and returns a distribution over the classes that sums to 1.

7. Rectified linear unit (ReLU)

$$\sigma(x) = \begin{cases} x & \text{if } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$

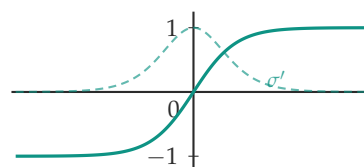
$$\frac{d}{dx}\sigma(x) = \begin{cases} 1 & \text{if } x > 0, \\ 0 & \text{otherwise.} \end{cases}$$



8. Tanh

$$\sigma(x) = \frac{1 - \exp\{-2x\}}{1 + \exp\{-2x\}}$$

$$\frac{d}{dx}\sigma(x) = 1 - \tanh^2(x)$$



S-shaped like sigmoid, but ranges from -1 to 1 . (See the notebook on activations.)

9. The hidden layer as a learned representation

$$\mathbf{x}_i \in \mathbb{R}^2$$

$$\mathbf{h}_i = \sigma(\mathbf{W}^h \mathbf{x}) \in \mathbb{R}^2$$

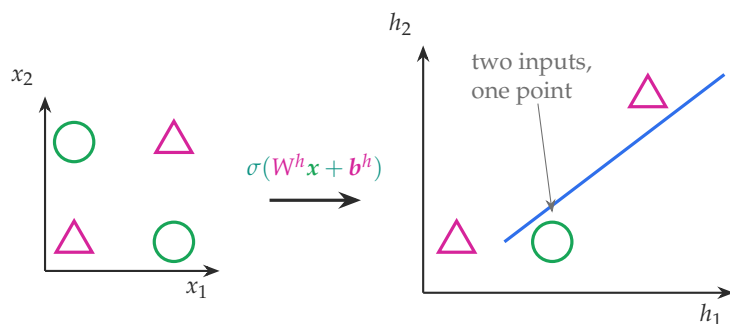
$$\hat{y}_i = \sigma(\underbrace{\mathbf{W}^y \cdot \mathbf{h}_i}_{\text{still a linear model! just over } \mathbf{h}})$$

still a linear model! just over $\mathbf{h}$

Suppose σ is ReLU and:

$$\mathbf{W}^h = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad \mathbf{b}^h = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Consider XOR again.



Running the four inputs through the hidden layer, $\mathbf{h}_i = \text{ReLU}(\mathbf{W}^h \mathbf{x}_i + \mathbf{b}^h)$:

$$\mathbf{x}_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \mathbf{h}_1 = \text{ReLU}\left(\begin{bmatrix} 0 \\ -1 \end{bmatrix}\right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \quad \mathbf{h}_2 = \text{ReLU}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{h}_3 = \text{ReLU}\left(\begin{bmatrix} 1 \\ 0 \end{bmatrix}\right) = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\mathbf{x}_4 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \mathbf{h}_4 = \text{ReLU}\left(\begin{bmatrix} 2 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$